



Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

Quantitative Social Research II

Workshop 8: Hierarchical Data

Jose Pina-Sánchez



Workshop Aims

Workshop Aims

Hierarchical Data

The Assumption of Independence

Adjustment Strategies

Multilevel Modelling

Recap

- Learn to identify hierarchical data
- Discuss the implications of violating the assumption of independence
- Modelling strategies considering lack of independence as a data nuisance problem to be adjusted
 - robust standard errors
- Modelling strategies considering lack of independence as a substantively interesting process to be modelled
 - multilevel modelling (aka hierarchical, random effects, or mixed effects models)



Workshop Aims: Recap

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- Assumptions in the linear regression model ($Y = \alpha + \beta_k X_k + e$):
 - normality: residuals are normally distributed
 - homoskedasticity: the variance of the residuals is constant
 - **independence**: residuals are independent of each other
 - no multicollinearity
 - perfectly measured variables
 - no missing data (other than missing at random)
 - no unobserved confounders: we control for all common causes of X_1 and Y
 - no reverse causality: Y does not cause X_1
 - linearity: the effect of X_1 on Y is the same across the range of X_1



Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

Hierarchical Data

- When cases composing a sample can be grouped within clusters
 - e.g. students within modules within programs
 - as a result of - or because you are - taking part in this module you share some commonalities (within-cluster correlation) that make you different from other students
 - additional within correlations could be expected from being enrolled in a Sociology/ Criminology program
 - and the same applies to other students in different modules and programs
- Question: can you think of any other examples of hierarchical data?



Hierarchical Data

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- When cases composing a sample can be grouped within clusters
 - e.g. students within modules within programs
 - as a result of - or because you are - taking part in this module you share some commonalities (within-cluster correlation) that make you different from other students
 - additional within correlations could be expected from being enrolled in a Sociology/ Criminology program
 - and the same applies to other students in different modules and programs
- Question: can you think of any other examples of hierarchical data?
 - interviewees within regions within countries
 - sentences imposed by judges sitting in courts
 - any instance where cluster sampling is used



Hierarchical Data

Workshop Aims

**Hierarchical
Data**

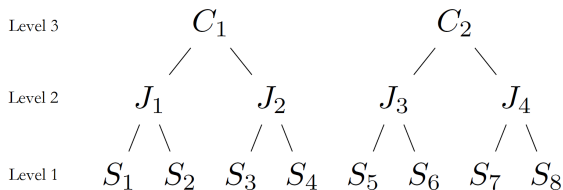
The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

The Hierarchical Structure of Sentencing Data





Hierarchical Data

Workshop Aims

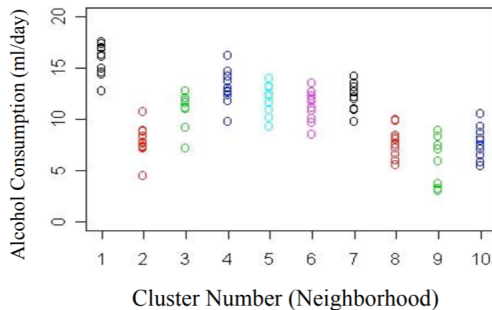
Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap



Source: Francesca Dominici

- Cases across this sample are not independent
- Cases within the same cluster are related to each other

Hierarchical Data: Notation

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- We need different subscripts to distinguish units at different levels
 - for the case of sentencing data we have considered three levels
 - court: $l = 1, 2, 3, \dots, L$
 - judge: $j = 1, 2, 3, \dots, J_l$
 - sentence: $i = 1, 2, 3, \dots, I_{lj}$
- We will use these to identify values in our outcome, explanatory variables and residuals

$$Y_{lji} = \beta_0 + \beta_k X_{klji} + \underbrace{e_{lji}}_{v_l + u_{lj} + \epsilon_{lji}}$$

- notice how the residual term can now be partitioned to reflect the unobserved variability stemming from each level



The Assumption of Independence

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- To estimate the standard errors of regression coefficients we use the variance covariance matrix
 - a matrix of the residuals' variances and covariances for each observation, for a simplified model of only $n = 3$ we have

$$\begin{pmatrix} \text{var}(e_1) & \text{cov}(e_1, e_2) & \text{cov}(e_1, e_3) \\ \text{cov}(e_2, e_1) & \text{var}(e_2) & \text{cov}(e_2, e_3) \\ \text{cov}(e_3, e_1) & \text{cov}(e_3, e_2) & \text{var}(e_3) \end{pmatrix}$$



The Assumption of Independence

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- To estimate the standard errors of regression coefficients we use the variance covariance matrix

- a matrix of the residuals' variances and covariances for each observation, for a simplified model of only $n = 3$ we have

$$\begin{pmatrix} \text{var}(e_1) & \text{cov}(e_1, e_2) & \text{cov}(e_1, e_3) \\ \text{cov}(e_2, e_1) & \text{var}(e_2) & \text{cov}(e_2, e_3) \\ \text{cov}(e_3, e_1) & \text{cov}(e_3, e_2) & \text{var}(e_3) \end{pmatrix}$$

- under the assumption of homoskedasticity and independence

$$\begin{pmatrix} \sigma_e^2 & 0 & 0 \\ 0 & \sigma_e^2 & 0 \\ 0 & 0 & \sigma_e^2 \end{pmatrix}$$

- under two level hierarchical data the diagonals will be equal to $\sigma_{\epsilon_i}^2 + \sigma_{u_j}^2$ and the 0s equal to σ_{u_j, u_j}

The Assumption of Independence

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- To estimate the standard errors of regression coefficients we use the variance covariance matrix

- a matrix of the residuals' variances and covariances for each observation, for a simplified model of only $n = 3$ we have

$$\begin{pmatrix} \text{var}(e_1) & \text{cov}(e_1, e_2) & \text{cov}(e_1, e_3) \\ \text{cov}(e_2, e_1) & \text{var}(e_2) & \text{cov}(e_2, e_3) \\ \text{cov}(e_3, e_1) & \text{cov}(e_3, e_2) & \text{var}(e_3) \end{pmatrix}$$

- under the assumption of homoskedasticity and independence

$$\begin{pmatrix} \sigma_e^2 & 0 & 0 \\ 0 & \sigma_e^2 & 0 \\ 0 & 0 & \sigma_e^2 \end{pmatrix}$$

- under two level hierarchical data the diagonals will be equal to $\sigma_{\epsilon_i}^2 + \sigma_{u_j}^2$ and the 0s equal to σ_{u_j, u_j}
- assuming independence in the presence of hierarchical data will lead to 'naive' findings
- underestimated measures of uncertainty (smaller SEs, narrower CIs, higher chance of type I errors)



Type I & II Errors

Workshop Aims

Hierarchical
Data

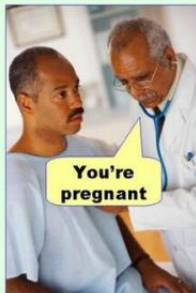
The Assumption
of Independence

Adjustment
Strategies

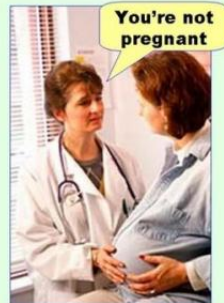
Multilevel
Modelling

Recap

Type I error
(false positive)



Type II error
(false negative)



Source: Paul Ellis 'Effect Sizes'



Strategies to Adjust for Within-Cluster Correlation

- Three main choices, all with pros and cons

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

**Adjustment
Strategies**

Multilevel
Modelling

Recap



Strategies to Adjust for Within-Cluster Correlation

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- Three main choices, all with pros and cons
- Robust standard errors (the ‘sandwich estimator’)
 - each variance and covariance is estimated empirically
$$e_{ji} = Y_{ji} - \beta_0 - \beta_k X_{kji}$$
 - pros: provides robust SEs
 - cons: within-cluster correlation is seen as a data nuisance, i.e. we do not model and learn about these correlations



Strategies to Adjust for Within-Cluster Correlation

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- Three main choices, all with pros and cons
- Robust standard errors (the ‘sandwich estimator’)
 - each variance and covariance is estimated empirically
$$e_{ji} = Y_{ji} - \beta_0 - \beta_k X_{kji}$$
 - pros: provides robust SEs
 - cons: within-cluster correlation is seen as a data nuisance, i.e. we do not model and learn about these correlations
- Fixed effects models
 - clusters are included in the model as dummy variables
$$Y_{ji} = \beta_0 + \beta_k X_{kji} + \beta_j X_{ji} + e_{ji}$$
 - pros: can model mean differences in the outcome by cluster which can be substantially interesting; e.g. which is the neighbourhood with higher alcohol consumption?
can control for cluster-level confounders; e.g. we might want to explore the effect of social class on alcohol consumption, which can be confounded by type of neighbourhood
 - cons: can lead to overfitted models



Strategies to Adjust for Within-Cluster Correlation

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- Multilevel modelling (MLM)

- the error term at each level is partitioned and modelled separately

$$Y_{ji} = \overbrace{\beta_0 + \beta_k X_{kji}}^{\text{fixed part}} + \overbrace{u_j + \epsilon_{ji}}^{\text{random part}}$$

- that's why MLM are often called mixed or random effects models, and why we called fixed effects models that way



Strategies to Adjust for Within-Cluster Correlation

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- Multilevel modelling (MLM)

- the error term at each level is partitioned and modelled separately

$$Y_{ji} = \overbrace{\beta_0 + \beta_k X_{kji}}^{\text{fixed part}} + \overbrace{u_j + \epsilon_{ji}}^{\text{random part}}$$

- that's why MLM are often called mixed or random effects models, and why we called fixed effects models that way
- pros: if modelled properly can provide robust SEs

Allows modelling variability between and within clusters:

e.g.1 Are there between court inconsistencies in sentencing?

e.g.2 Are differences in happiness due to differences across countries or individuals?



Strategies to Adjust for Within-Cluster Correlation

Workshop Aims

Hierarchical
DataThe Assumption
of IndependenceAdjustment
StrategiesMultilevel
Modelling

Recap

- Multilevel modelling (MLM)

- the error term at each level is partitioned and modelled separately

$$Y_{ji} = \underbrace{\beta_0 + \beta_k X_{kji}}_{\text{fixed part}} + \underbrace{u_j + \epsilon_{ji}}_{\text{random part}}$$

- that's why MLM are often called mixed or random effects models, and why we called fixed effects models that way

- pros: if modelled properly can provide robust SEs

Allows modelling variability between and within clusters:

e.g.1 Are there between court inconsistencies in sentencing?

e.g.2 Are differences in happiness due to differences across countries or individuals?

- cons: don't control for cluster-level confounders

invoke further assumptions:

$$u_j \sim N(0, \sigma_u) ; \text{cov}(u_j, u_{j'}) = 0$$

$$\epsilon_j \sim N(0, \sigma_\epsilon) ; \text{cov}(\epsilon_j, \epsilon_{j'}) = 0$$

$$\text{cov}(\epsilon_{ji}, u_j) = 0$$



Random Intercepts

- The simplest form of MLM
 - allows for the intercept to vary across clusters
 - for the case of a 2-level MLM with one explanatory variable could be expressed as

$$Y_{ji} = \overbrace{\beta_0}^{\beta_0 + u_j} + \beta_1 X_{1ji} + \epsilon_{ji}$$

- invokes the same assumptions listed in the previous slide



Random Intercepts

- The simplest form of MLM
 - allows for the intercept to vary across clusters
 - for the case of a 2-level MLM with one explanatory variable could be expressed as

$$Y_{ji} = \overbrace{\beta_0}^{\beta_0 + u_j} + \beta_1 X_{1ji} + \epsilon_{ji}$$

- invokes the same assumptions listed in the previous slide
- Can be used to estimate the intraclass correlation coefficient
 - $ICC = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\epsilon^2}$
 - the proportion of unobserved variability in the outcome variable (i.e. residual variability) stemming from level 2, e.g. the proportion of sentencing disparities due to between judge differences
 - can also be understood as the correlation between observations from the same cluster, e.g. the similarities between sentences imposed by the same judge



Random Intercepts

- The simplest form of MLM
 - allows for the intercept to vary across clusters
 - for the case of a 2-level MLM with one explanatory variable could be expressed as

$$Y_{ji} = \overbrace{\beta_0}^{\beta_0 + u_j} + \beta_1 X_{1ji} + \epsilon_{ji}$$

- invokes the same assumptions listed in the previous slide
- Can be used to estimate the intraclass correlation coefficient
 - $ICC = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\epsilon^2}$
 - the proportion of unobserved variability in the outcome variable (i.e. residual variability) stemming from level 2, e.g. the proportion of sentencing disparities due to between judge differences
 - can also be understood as the correlation between observations from the same cluster, e.g. the similarities between sentences imposed by the same judge
- Can be extended to 3 or more levels

$$Y_{lji} = \overbrace{\beta_0}^{\beta_0 + v_l + u_{lj}} + \beta_1 X_{1lji} + \epsilon_{lji}$$

Workshop Aims

Hierarchical
DataThe Assumption
of IndependenceAdjustment
StrategiesMultilevel
Modelling

Recap



Random Intercepts

Workshop Aims

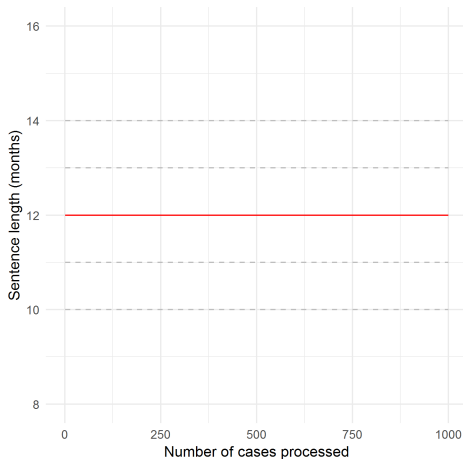
Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

**Multilevel
Modelling**

Recap





Random Intercepts

Workshop Aims

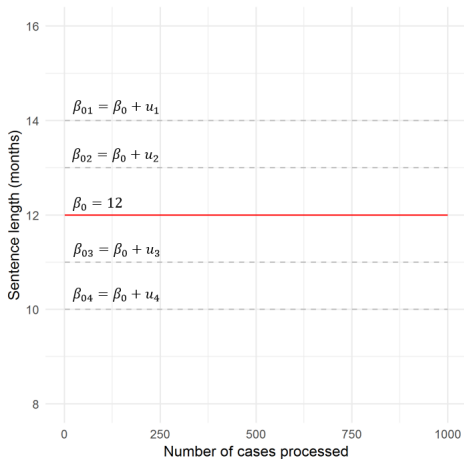
Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap





Random Slopes

- The RIs model can be extended by allowing between cluster variability around the intercept but also around specific slopes

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

**Multilevel
Modelling**

Recap



Random Slopes

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- The RIs model can be extended by allowing between cluster variability around the intercept but also around specific slopes
 - for the case of a 2-level MLM with one explanatory variable could be expressed as

$$Y_{ji} = \underbrace{\beta_0 + u_{0j}}_{\beta_{0j}} + \underbrace{\beta_1 + u_{1j}}_{\beta_{1j}} X_{1ji} + \epsilon_{ji}$$



Random Slopes

Workshop Aims

Hierarchical
DataThe Assumption
of IndependenceAdjustment
StrategiesMultilevel
Modelling

Recap

- The RIs model can be extended by allowing between cluster variability around the intercept but also around specific slopes

- for the case of a 2-level MLM with one explanatory variable could be expressed as

$$Y_{ji} = \underbrace{\beta_0 + u_{0j}}_{\beta_{0j}} + \underbrace{\beta_1 + u_{1j}}_{\beta_{1j}} X_{1ji} + \epsilon_{ji}$$

- as before, level-1 and level-2 residuals are assumed to be

$$u_{0j} \sim N(0, \sigma_{u0}) ; cov(u_{0j}, u_{0j'}) = 0$$

$$u_{1j} \sim N(0, \sigma_{u1}) ; cov(u_{1j}, u_{1j'}) = 0$$

$$\epsilon_j \sim N(0, \sigma_\epsilon) ; cov(\epsilon_j, \epsilon_{j'}) = 0$$



Random Slopes

Workshop Aims

Hierarchical
DataThe Assumption
of IndependenceAdjustment
StrategiesMultilevel
Modelling

Recap

- The RIs model can be extended by allowing between cluster variability around the intercept but also around specific slopes

- for the case of a 2-level MLM with one explanatory variable could be expressed as

$$Y_{ji} = \underbrace{\beta_{0j}}^{\beta_0 + u_{0j}} + \underbrace{\beta_{1j}}^{\beta_1 + u_{1j}} X_{1ji} + \epsilon_{ji}$$

- as before, level-1 and level-2 residuals are assumed to be

$$u_{0j} \sim N(0, \sigma_{u0}) ; cov(u_{0j}, u_{0j'}) = 0$$

$$u_{1j} \sim N(0, \sigma_{u1}) ; cov(u_{1j}, u_{1j'}) = 0$$

$$\epsilon_j \sim N(0, \sigma_\epsilon) ; cov(\epsilon_j, \epsilon_{j'}) = 0$$

- however, now we might be interested in exploring whether $cov(u_{0j}, u_{1j'}) \neq 0$
- if positive the slopes will diverge, i.e. higher intercepts are associated with higher slopes and vice versa
- if negative the slopes will converge, i.e. higher intercepts are associated with lower slopes and vice versa



Workshop Aims

Hierarchical
Data

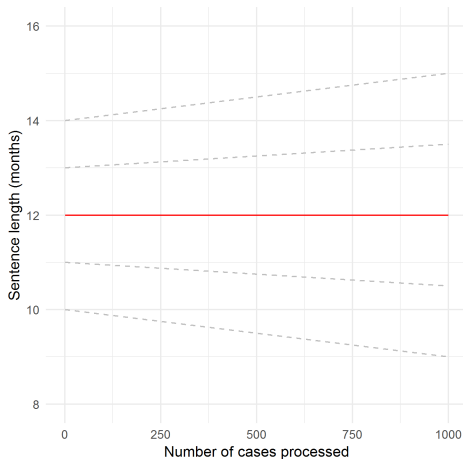
The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

Random Slopes (+cov)





Random Slopes (+cov)

Workshop Aims

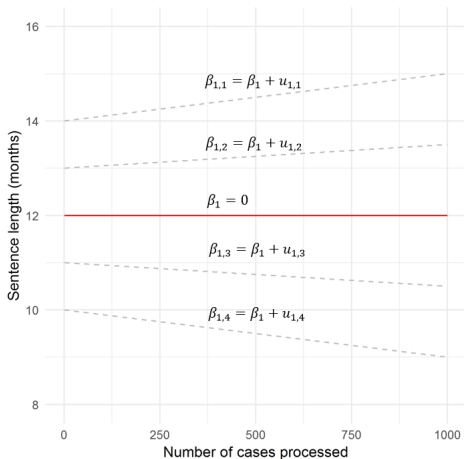
Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap





Workshop Aims

Hierarchical
Data

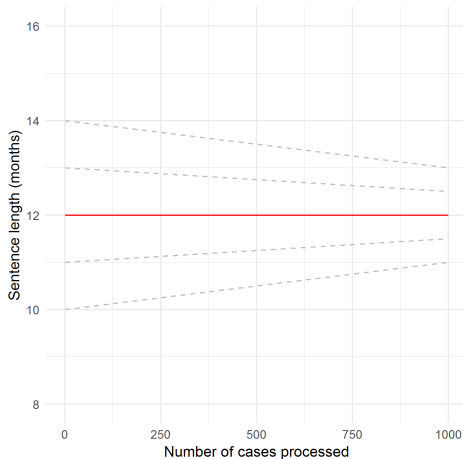
The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

Random Slopes (-cov)





Recap

- In the presence of hierarchical data the assumption of independence does not hold
 - measures of uncertainty will tend to be underestimated $\rightarrow$ type I errors

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap



Recap

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- In the presence of hierarchical data the assumption of independence does not hold
 - measures of uncertainty will tend to be underestimated $\rightarrow$ type I errors
- We have covered the three main adjustment strategies
 - robust standard errors
 - fixed effects
 - multilevel modelling



Recap

Workshop Aims

Hierarchical
DataThe Assumption
of IndependenceAdjustment
StrategiesMultilevel
Modelling

Recap

- In the presence of hierarchical data the assumption of independence does not hold
 - measures of uncertainty will tend to be underestimated → type I errors
- We have covered the three main adjustment strategies
 - robust standard errors
 - fixed effects
 - multilevel modelling
- Robust standard errors (the ‘sandwich estimator’)
 - provide unbiased measures of uncertainty (also in the presence of heteroskedasticity)
 - doesn’t control for systematic difference between clusters (potential confounders)
 - doesn’t tell us anything about between/within cluster variability
 - to be used when cluster variability is not of interest (considered a data nuisance)



Recap

- Fixed effects
 - partially adjust SEs while controlling for systematic differences between clusters
 - to be used when confounders are a serious concern
 - can be used to compare means across clusters but not great at assessing variability
 - if the number of clusters is large will risk overfitting the model

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap



Recap

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- Fixed effects
 - partially adjust SEs while controlling for systematic differences between clusters
 - to be used when confounders are a serious concern
 - can be used to compare means across clusters but not great at assessing variability
 - if the number of clusters is large will risk overfitting the model
- Multilevel modelling
 - more flexible than fixed effects models at adjusting SEs
 - does not control for systematic differences between clusters
 - allow exploring substantive questions related to between/within cluster variability

Recap

Workshop Aims

Hierarchical
Data

The Assumption
of Independence

Adjustment
Strategies

Multilevel
Modelling

Recap

- Fixed effects
 - partially adjust SEs while controlling for systematic differences between clusters
 - to be used when confounders are a serious concern
 - can be used to compare means across clusters but not great at assessing variability
 - if the number of clusters is large will risk overfitting the model
- Multilevel modelling
 - more flexible than fixed effects models at adjusting SEs
 - does not control for systematic differences between clusters
 - allow exploring substantive questions related to between/within cluster variability
- To learn more about multilevel modelling
 - read Goldstein (1995) Chapter 2
 - and watch the online course from Brunton-Smith (2019)
 - sign up for the LEMMA online course